

Paper Manager pack — LeWorldModel / LeWM (Dec 2 · Accompanying Models)

Paper: LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels
arXiv: <https://arxiv.org/abs/2603.19312> · [code](#) · [site](#) · arXiv **2603.19312v3** (3 Jun 2026)
Authors: Lucas Maes*¹, Quentin Le Lidec*², Damien Scieur^{1,3}, Yann LeCun², Randall Balestriero⁴ (*equal) — Mila & UdeM / NYU / Samsung SAIL / Brown
PDF: </workspace/papers/leworldmodel-2603.19312.pdf> (= </workspace/leworldmodel-quiet/paper.pdf>) · **28 pp** · ~5.4 MB
Session: Dec 2 · due 9:30 AM · Type: **Accompanying Models (World Models, Reasoning, Planning)**
Role weights: Empiricist=3 (Nykolas Rekasius) **LONGEST PRIMARY**; Archaeologist / Stakeholder / Reviewer / Visionary seats empty — **still all five produced**; Empiricist heaviest
Kanta role: **NOT seated** this session — still synthesize Empiricist-richest five-role pack for Nykolas; deepen reproducible claims, planning speed/stability, training recipe without fragile heuristics; WM/planning + taxonomy deep dive; course-smoke run pack
Themes: compact ~15M E2E JEPA WM from pixels · **SIGReg** anti-collapse · drops EMA / stop-grad / frozen encoder / multi-term VICReg · ~48× faster planning than DINO-WM · **SIM-ONLY** (Push-T / OGBench-Cube / Two-Room / Reacher) — **not** a Meta V-JEPA descendant · sibling recipe **PLDM** → **LeWM** vs Meta **I-JEPA** → ... → **2-AC**
Sources: Summarizer (</workspace/papers/leworldmodel-2603.19312-summary.md>) + Senior Research (</workspace/leworldmodel-quiet/DELIVERABLE.md>) + Ablation Bot (</workspace/papers/leworldmodel-2603.19312-ablations.md> — **AUTHORITATIVE**) + [paper_notes.md](#) / [env_outline/](#)
Sibling links: </workspace/ijepa-quiet/> (heavy JEPA lineage) · </workspace/vjepa21-quiet/> (Meta dense/AC foil) · </workspace/siglip-quiet/> · </workspace/pi05-quiet/> · </workspace/openpi-comet-smoke/> · </workspace/diffusion-policy-quiet/>

1. One-liner

Stable end-to-end JEPA world model from raw pixels — next-embedding **MSE + SIGReg** only (~15M params, 1 GPU / few hours); drops EMA / stop-grad / frozen DINOv2 / PLDM's 7-term VICReg. Compact [CLS] latent → CEM planning **0.98 s vs DINO-WM 47 s** (~48×); Push-T **96.0±2.83** vs PLDM **78.0±5.0** / DINO **92.0±1.63**; fixed-FLOP Push-T **90 vs 13**. **Sim-only** — not Meta V-JEPA 2-AC.
Contrast poles (consistency triangle + control-WM branch):

Pole	What they do	Primary claim / use
DINOv2 / SigLIP	Discriminative JEA / dense image features	VLA towers; DINO-WM freeze
MAE / Dreamer / IRIS	Generative pixels / reward-tied WMs	FT or RL imagination
I-JEPA → V-JEPA → 2 → 2.1 → 2-AC	Meta predictive SSL → AC planning	Real-robot CEM (Franka) — parallel line
PLDM → LeWM	Compact E2E JEPA-WM from pixels	Fast sim latent MPC; this paper

Do NOT invent / claim: Franka / BEHAVIOR / DROID numbers (none in paper); LeWM as Meta V-JEPA descendant; BEHAVIOR hybrid JEPA+FM as in-paper result; course smoke = Push-T **96% / 48×** exact; “zero heuristics” while ignoring predictor dropout **0.1** + AdaLN zero-init + BN projector (allowed stabilizers).

2. Summary (exec overview + key numbers)

Maes & Le Lidec et al. (arXiv 2603.19312) attack fragile JEPA training for **action-conditioned latent world models**: EMA target encoders, stop-grad, frozen foundation towers, or PLDM's six+ VICReg-style loss weights. Fix = **two-term** objective ($L = L_{\text{pred}} + \lambda \cdot \text{SIGReg}(Z)$) (MSE next-embedding + Sketched-Isotropic-

Gaussian Regularizer from **LeJEPA**). Architecture: ViT-Tiny encoder (~5M) + AdaLN transformer predictor (~10M) ≈ **15M**; joint E2E from RGB; image-goal **CEM+MPC** in compact latent.

Why Accompanying Models session: Completes the JEPA lecture arc after I-JEPA (idea) and V-JEPA 2.1 / 2-AC (scale + real robot): **how to train a tiny AC WM without brittle heuristics**, with a spectacular systems hook (**0.98 s vs 47 s**). Placement = **sibling / alternative training recipe** in the control-domain E2E JEPA-WM niche — **not** a descendant of V-JEPA 2-AC.

Axis	Number / claim
Venue / code	arXiv 2603.19312v3 ; <code>lucas-maes/le-wm</code> + <code>le-wm.github.io</code> CLEAR
Params / train	~ 15M ; single GPU, few hours; 10 epochs typical; bs 128 ; 224 ² ; frame-skip 5
Loss	(L_{pred}) MSE + λ SIGReg; default $\lambda=0.1$, $M=1024$; tunable HPs 6 → 1 vs PLDM
Push-T SR (Tab. 5, 3 seeds)	LeWM 96.0±2.83 ; DINO-WM 92.0±1.63 ; PLDM 78.0±5.0 (~+18 pp vs PLDM)
Planning latency (Fig. 3)	LeWM 0.98 s vs DINO-WM 47 s (~48×); ~200× fewer tokens
Fixed-FLOP SR (Fig. 3)	Push-T 90 vs 13 ; OGBench-Cube 74 vs 48 (LeWM vs DINO-WM)
Recon ablation (Tab. 7)	w/o 96.0±2.83 → with 86.0±7.54 (recon hurts)
λ band (Fig. 16)	SR >80% for $\lambda \in [0.01, 0.2]$; peak ~0.09; sharp drop at $\lambda=0.5$
Dropout (Tab. 9)	0.0 → 78 ; 0.1 → 96.0 ; 0.2 → 85.33; 0.5 → 66.67
Predictor size (Tab. 6)	tiny 80.67 ; small 96.0 ; base 86.7
Encoder (Tab. 8)	ViT 96.0 ≈ ResNet-18 94.0
Solver (Tab. 10)	CEM 96 ; Adam 84 ; RMSProp 67 ; SGD 26 (LeWM Push-T)
Two-Room caveat	PLDM/DINO > LeWM (~87 vs ~97–100) — SIGReg vs low-intrinsic-dim
Scope	Sim-only Push-T / Cube / TwoRoom / Reacher — no Franka numbers

Must-cite tattoo: Push-T **96.0±2.83** · latency **0.98 vs 47** (~48×) · fixed-FLOP **90 vs 13** · recon **96** → **86** · $\lambda \in [0.01, 0.2]$ / dies at 0.5 · **TwoRoom caveat** · **no Franka**.

Do NOT claim: real-robot transfer; course smoke = paper absolutes; LeWM replaces V-JEPA 2-AC; BEHAVIOR hybrid as demonstrated; “first E2E JEPA” without fencing vs PLDM definitional boundary.

3. Keywords

LeWorldModel; LeWM; JEPA; SIGReg; Sketched-Isotropic-Gaussian Regularizer; end-to-end world model; latent planning; CEM; MPC; representation collapse; PLDM; DINO-WM; Push-T; OGBench-Cube; Two-Room; Reacher; reward-free offline; physical probing; violation-of-expectation; LeJEPA; Accompanying Models; BEHAVIOR hybrid (TENTATIVE)

4. Deep dive — World-model / planning + taxonomy (PRIMARY SESSION CONTRIBUTION)

Callout: Scientific product of *this* PDF = a **stable, compact, action-conditioned JEPA world model from pixels** that makes latent CEM **sub-second**. Session type = Accompanying Models because students need the **training-stability + planning-latency** half of the JEPA → control story after Meta’s scale/Franka half.

Taxonomy (Fig. 2 moral — three-way map)

Family	Collapse / objective trick	Latent	Planning	Foil role
Foundation-freeze WM (DINO-WM)	Freeze DINOv2	Dense patch tokens (~200×)	CEM; 47 s	Speed foil
E2E multi-term JEPA-WM (PLDM)	7-term VICReg-style	Compact	CEM; speed ≈ LeWM	Fragility foil
E2E two-term JEPA-WM (LeWM)	SIGReg only	Compact [CLS] ~192-d	CEM; 0.98 s	This inventum
Generative / reward WMs (Dreamer, TD-MPC, IRIS)	Pixel recon / reward	Pixels or states	RL imagination	Task-tied foil
Meta SSL → AC (V-JEPA 2 / 2.1 / 2-AC)	EMA+SG in SSL; AC post-train	Video / AC tokens	Franka CEM (CLEAR prior pack)	Parallel line — different domain

What is “world-model-ish” here — CLEAR (in-paper)

Piece	In-paper role	World-model reading
Encoder ($z_t = \text{enc}_{\theta}(o_t)$)	ViT-Tiny → [CLS] → MLP+BN	Pixels → compact state
Predictor ($\hat{z}_{t+1} = \text{pred}_{\phi}(z_t, a_t)$)	6L AdaLN TF; dropout 0.1	Action-conditioned dynamics ($P(z'$
Loss ($L_{\text{pred}} + \lambda \text{SIGReg}$)	MSE next-emb + isotropic Gaussian	Dynamics + anti-collapse without EMA/SG
CEM+MPC (§3.2)	H=5; pop=300; cost ($ \hat{z}_H - z_g _2^2$)	Image-goal latent control
Speed Fig. 3	0.98 s vs 47 s; fixed-FLOP 90 vs 13	Deploy hook = token compression
Probes + VoE	Tab. 1 / Fig. 8	Latent encodes physics; teleport surprise ($p < 0.01$)

What is not overclaimable

- **Sim-only** continuous control — Push-T / OGBench-Cube / Two-Room / Reacher. **No Franka / BEHAVIOR / DROID / Bridge numbers.**
- Two-Room: LeWM **underperforms** PLDM/DINO — SIGReg Gaussian prior vs low-complexity data (admitted).
- Cube: DINO-WM slightly ahead (~86 vs ~74).
- Speedup is largely **cheaper dynamics steps** (compact (z)), not fewer CEM iters — report solver-matched.
- Dropout **0.1** is an allowed regularizer (78 → 96), not “zero heuristics.”
- Hybrid LeWM imagination + $\pi_{0.5}$ /FM for BEHAVIOR = **Visionary TENTATIVE**, not in-paper.
- LeWM is not a Meta V-JEPA descendant — sibling **control E2E** recipe (PLDM → LeWM).

SIBLING recipe (session tattoo — do not collapse lines)

Meta SSL JEPA:	I-JEPA → V-JEPA → V-JEPA 2 → 2.1 → 2-AC (Franka CEM) [packs: ijepa-quiet / vjepa21-quiet]
Control E2E JEPA-WM:	PLDM (fragile multi-term VICReg) → **LeWM (MSE+SIGReg)** ← THIS PAPER
Foundation freeze:	DINO-WM (DINOv2 frozen + predictor; slow tokens)

World-model exam bite: "LeWM = compact ~15M E2E JEPA WM from pixels: MSE+SIGReg, no EMA/SG/freeze/multi-VICReg; Push-T **96.0±2.83**; plan **0.98 vs 47 s** (~48×); fixed-FLOP **90 vs 13**; recon hurts **96 → 86**; $\lambda \in [0.01, 0.2]$ / dies at 0.5; **TwoRoom caveat**; **sim-only — no Franka**. Sibling **PLDM → LeWM**, not Meta 2-AC child."

5. Deep dive — SIGReg / heuristics-off recipe (Empiricist spine methods)

Callout: App. G + Figs. 15–16 / Tabs. 5–10 are the session’s methods spine. Fragile multi-term / EMA JEPA → two-term SIGReg → stable curves + one HP + fast latent CEM.

Architecture checklist (Empiricist pin)

Piece	Spec (paper default)
Encoder	ViT-Tiny /14, 12L, 3H, dim 192 (~5M); alt ResNet-18
Projector (enc)	1-layer MLP + BatchNorm (final LN blocks SIGReg)
Predictor	Transformer ~ViT-S, 6L, 16H , dropout 0.1 (~10M); AdaLN actions (zero-init)
Projector (pred)	Twin of enc projector
History	N=3 (PushT/Cube); N=1 (TwoRoom); causal AR
Losses	$(L_{\text{pred}} = \ \hat{z}_{t+1} - z_{t+1} \ ^2)$; SIGReg = M random projections + Epps–Pulley
Defaults	M=1024 , $\lambda=0.1$; batch 128 ; sub-traj 4 ; frame-skip 5 ; 224 ²
Planner	CEM 300 samples; 30 iters PushT / 10 else; elite 30; H=5 (=25 env steps)
Total	~15M; 10 epochs; reward-free offline (o,a) only
Forbidden in recipe	EMA target, stop-grad, frozen pretrained enc, multi-term VICReg, pixel recon (unless ablating)

Ablation Bot AUTHORITATIVE — must-hit deltas

Ablation	Paper locus	Headline numbers
Push-T multi-seed	Tab. 5	LeWM 96.0±2.83 ; PLDM 78.0±5.0 ; DINO 92.0±1.63
Planning speed / FLOPs	Fig. 3	0.98 s vs 47 s (~48×); fixed-FLOP Push-T 90 vs 13 ; Cube 74 vs 48
λ SIGReg	Fig. 16	>80% for $\lambda \in [0.01, 0.2]$; peak ~0.09; dies at 0.5
M / knots	Fig. 15	Largely <i>insensitive</i> — λ is the real HP
Emb dim	Fig. 15	Drop below ~184; saturate above
Decoder / recon	Tab. 7	w/o 96.0 vs with 86.0±7.54
Predictor dropout	Tab. 9	0 → 78 ; 0.1 → 96 ; 0.2 → 85; 0.5 → 67
Predictor size	Tab. 6	tiny 80.67 ; small 96.0 ; base 86.7
Encoder arch	Tab. 8	ViT 96.0 ≈ RN18 94.0
Solver	Tab. 10	CEM 96 / Adam 84 / RMSProp 67 / SGD 26
Two-Room / Cube	Fig. 6	TwoRoom LeWM behind; Cube DINO slightly ahead
Not factorial	§3.1	EMA/SG/freeze claimed removed — no within-LeWM add-back table

Taxonomy / recipe exam bite: "LeWM drops EMA/SG/freeze/multi-VICReg for MSE+SIGReg (λ -only); Push-T **96±2.83**; recon hurts **96** → **86**; λ band [0.01,0.2] / cliff at 0.5; dropout 0.1 allowed; CEM $\gg$ SGD; **sim-only**."

6. Role-ready sections (all five — Empiricist RICHEST)

6.1 Empiricist — WEIGHT 3 LONGEST PRIMARY (Nykolas Rekasius)

Seat charge: Deepen reproducible claims, planning speed/stability, and the heuristics-off training recipe. Longest talk block (~16–18 slides). Course smokes that stress **SIGReg E2E · λ cliff · planning wall-time · lite baselines**, ranked by GPU honesty. Prefer PushT-lite / synthetic + ViT-Tiny — **not** full 4-env 10-epoch / foundation DINO bake-off. **Kanta not seated** — Nykolas carries Empiricist.

Claim under measurement

Matched short pretrain: **SIGReg-only E2E** trains without collapse while **no-reg** collapses; mid- λ planning $\gg \lambda \in \{0, 0.5\}$; compact latent CEM $\ll$ high-token DINO stub wall-time; lewm-smoke $\geq$ pldm_lite on PushT-proxy — **directional**, not paper **96% / 48×**.

Hypotheses (falsifiable — from DELIVERABLE)

1. **H-stable (#1)**: SIGReg-E2E (no SG/EMA/freeze) trains without collapse; no-reg collapses (latent std/rank).
2. **H-fragile (#1)**: EMA+SG unnecessary under SIGReg; VICReg-lite shows noisier multi-loss / higher variance (Tab. 5 moral).
3. **H-lambda (#2)**: Planning proxy high for $\lambda \in \{0.01, 0.1\}$; crashes/dynamics-hurt at 0.5; $\lambda=0$ fails (Fig. 16).
4. **H-speed (#3)**: Compact CLS latent plans $\gg$ faster wall-clock than high-token DINO stub at matched CEM; at fixed wall-time, compact $\geq$ stub proxy SR.
5. **H-rank (#4)**: lewm-smoke $\geq$ pldm_lite; competitive with dino_wm_lite — **not** require 96/78/92.
6. **H-decode (#5)**: Pixel recon aux hurts or flats planning proxy (Tab. 7 96 → 86).
7. **H-drop (#6)**: Dropout $p=0.1 > 0$ and $\gg 0.5$ (Tab. 9).
8. **H-pred-size (#7)**: ViT-S $\geq$ ViT-T; ViT-B not required (Tab. 6).
9. **H-phys (#12)**: Probes recover pose; VoE teleport surprise $>$ color (Fig. 8).
10. **H-lineage**: Logs must mark LeWM = SIGReg E2E (no EMA target) vs I-JEPA/V-JEPA EMA lineage + AC MPC.

Falsifiers: no-reg does not collapse; $\lambda=0.5 \geq \lambda=0.1$; DINO-stub faster at matched tokens; recon raises SR; dropout 0 ≥ 0.1 ; smoke claimed as Push-T 96% / 48× / Franka.

Prioritized ablation matrix (Ablation Bot AUTHORITATIVE + DELIVERABLE)

#	Experiment	Paper hook	Smoke	Pri
RC1	Heuristics off: SIGReg E2E vs noreg / ema_sg / vicreg_lite	§3; Fig. 18–19; Tab. 5	Y — PRIMARY	P0
RC2	λ sweep {0, 0.01, 0.1, 0.5} (+ optional M)	Fig. 16 / 15	Y — PRIMARY	P0
RC3	Planning speed / fixed wall-time vs DINO stub	Fig. 3	Y — PRIMARY	P0
RC4	vs PLDM-lite / DINO-WM-lite on PushT-lite	Fig. 6 / Tab. 5	Y / Partial	P0
RC5	Decoder / recon on vs off	Tab. 7	Y	P1
RC6a	Predictor dropout {0, 0.1, 0.5}	Tab. 9	Y	P1
RC6b	Predictor tiny vs small	Tab. 6	Partial	P2
RC2b	SIGReg M {64, 1024}	Fig. 15	Y cheap	P1
RC3b	Solver CEM vs Adam	Tab. 10	Partial	P1
RC7	Physical probe + VoE stub	Tab. 1 / Fig. 8	Partial	P2
—	Full 4-env 10-ep; foundation DINO; exact Tab. 5 3-seed; TwoRoom paper protocol	—	N	Skip

Min package: RC1 + RC2 + RC3 + RC4 → RC5 → RC6a → optional RC2b/RC3b/RC7.

Exact reproduce stack (EDIT)

```
# Official
git clone https://github.com/lucas-maes/le-wm
# Pin commit → logs/pins.txt ; site https://le-wm.github.io

# Course scout env (quiet outlines – preferred)
export LEWM_WORK_ROOT=$SCRATCH/lewm-smoke
cd /workspace/leworldmodel-quiet
bash env_outline/setup_mamba.sh
# mamba activate lewm-scout
mkdir -p logs runs checkpoints configs
# EDIT partition/account when fleshing slurm_*.sh from DELIVERABLE §3 cards
# Outlines only – do NOT sbatch from quiet box
# NEVER claim paper Push-T 96% / 0.98s / 48× / fixed-FLOP 90 from smoke
# NEVER invent Franka / BEHAVIOR numbers
```

Run cards (from DELIVERABLE — Empiricist PRIMARY)

RC1 — heuristics_off_smoke (#1 — PRIMARY)

Field	Spec
Paper	No SG/EMA/pretrained; ($L=L_{\mathrm{pred}}+\lambda\mathrm{SIGReg}$); vs PLDM 7-term / JEPA EMA+SG / DINO freeze
Goal	SIGReg-E2E trains + plans without collapse; noreg collapses
Train	ViT-Tiny enc + ViT-S pred (or smaller); 1 seed; 2k–15k steps on tiny shard; $\lambda=0.1$
Axes	recipe $\in \{\text{sigreg_e2e}, \text{noreg}, \text{ema_sg_jepa}, \text{vicreg_lite}\}$
Metrics	(L_{pred}), SIGReg; latent std / PCA rank; open-loop MSE; optional CEM proxy SR
Pass	≥ 2 recipes finish; sigreg non-collapse; noreg collapse or worse proxy — not paper 96%
Budget	~4–12 GPU-h · VRAM 10–20 GB
Script	env_outline/slurm_heuristics_off_smoke.sh (named in DELIVERABLE; flesh from RC1 card)

RC2 — lambda_sweep (#2 — PRIMARY)

Field	Spec
Paper	Fig. 16; $\lambda \in [0.01, 0.2]$ SR>80%; peak ~0.09; $\lambda=0.5$ hurts
Grid	$\lambda \in \{0.0, 0.01, 0.1, 0.5\}$; optional $M \in \{64, 1024\}$ at $\lambda=0.1$
Pass	≥ 3 λ points; mid > extremes on planning proxy or latent health+rollout MSE
Budget	~3–10 GPU-h
Script	env_outline/slurm_lambda_sweep.sh

RC3 — planning_speed_bench (#3 — PRIMARY)

Field	Spec
Paper	Fig. 3: 47 s → 0.98 s; ~48×; fixed-FLOP 90 vs 13
Goal	encode+CEM wall-time + SR vs high-token DINO stub under same CEM (pop=300, iters≤10–30, H=5)
Pass	Timing table; speedup ratio $\gg 1$ vs DINO-stub; deterministic seed
Budget	≤1–4 GPU-h (infer)
Script	env_outline/slurm_planning_speed_bench.sh

RC4 — vs_jepa_wm_baselines (#4 — PRIMARY)

Field	Spec
Paper	Tab. 5 / Fig. 6: 96 > 92 > 78; +18 pp vs PLDM; TwoRoom caveat
Train	Short matched {lewm_smoke, pldm_lite, dino_wm_lite}; pixels-only
Pass	lewm ≥ pldm_lite on proxy; competitive with dino_lite; 1-seed variance note
Budget	~6–16 GPU-h
Script	env_outline/slurm_vs_jepa_wm_baselines.sh

RC5 / RC6 / RC7 — follow-ons

RC5 decoder on/off (Tab. 7 moral); RC6a dropout {0,0.1,0.5} ; RC6b tiny vs small; RC7 VoE/probe stub — after P0 green.

Numbers to memorize (exam)

Fact	Value
Push-T Tab. 5	LeWM 96.0±2.83 · DINO 92.0±1.63 · PLDM 78.0±5.0
Latency Fig. 3	0.98 s vs 47 s (~48×)
Fixed-FLOP Push-T / Cube	90 vs 13 · 74 vs 48
Recon Tab. 7	96.0 → 86.0
λ Fig. 16	[0.01, 0.2] >80%; peak ~0.09; cliff at 0.5
Dropout Tab. 9	0 → 78 / 0.1 → 96 / 0.5 → 67
Pred size Tab. 6	tiny 80.67 / small 96.0 / base 86.7
Encoder Tab. 8	ViT 96 / RN18 94
Solver Tab. 10	CEM 96 / Adam 84 / SGD 26
Params / HP	~15M · HPs 6 → 1 · O(log n) vs O(n ⁶)
Scope	Sim-only · no Franka · TwoRoom caveat

Failure modes cheat-sheet

Symptom	Likely cause	Next action
All recipes collapse	λ too high / projector BN missing	λ=0.1; add BN after LN
noreg does not collapse	Accidental SIGReg left on	Audit loss; log recipe flag
λ sweep flat	Steps too short / weak eval	Longer train; use CEM proxy not only (L_{pred})
No speedup vs DINO	Token counts matched accidentally	Log #tokens ; patch tokens for DINO stub
PLDM-lite ≈ LeWM	VICReg-lite not multi-term	Port ≥3–7 terms or label “not PLDM”
CEM SR noise	Goals unreachable / H too short	Match H=5 / skip=5; filter reachable
OOM at 224 ² bs128	Smoke too close to paper	128 ² , bs32–64, ViT-T
“Heuristics off” but EMA in code	Copied I-JEPA trainer	Delete target EMA; single encoder
Claimed 96% / 48× / Franka from smoke	Overclaim	Report directional only

What not to invent / claim

- Do not claim course smoke = Push-T 96% / latency 0.98 s / 48× / fixed-FLOP 90.
- Do not invent Franka / BEHAVIOR / DROID numbers.
- Do not launch full 4-env 10-ep / foundation DINO bake-off from course queue.
- Do not call dropout “zero heuristics.”

- Do not present LeWM as Meta V-JEPA 2-AC replacement.

Empiricist exam bite: "Maes'26 LeWM: MSE+SIGReg E2E (~15M); Push-T **96.0±2.83**; **0.98 vs 47 s**; fixed-FLOP **90 vs 13**; recon **96 → 86**; $\lambda \in [0.01, 0.2]$ /cliff 0.5; Empiricist RC1+RC2+RC3+RC4 on PushT-lite for **direction** — never claim 96%/48×/Franka from smoke."

6.2 Archaeologist — produced despite empty seat

Seat charge: Place LeWM on the JEPA map without rewriting I-JEPA / V-JEPA 2.1 CLEAR labels. Emphasize SIBLING control recipe (PLDM → LeWM) vs Meta (I-JEPA → ... → 2-AC). Fence sim-only.

Priors — CLEAR (cited / direct)

Prior	Role relative to LeWM	Label
LeCun 2022 AMI	Names JEPA	CLEAR
I-JEPA Assran'23	Emb prediction; EMA/SG	CLEAR (SSL parent idea)
V-JEPA / V-JEPA 2	Video JEPA; scale; planning mention	CLEAR (cited; different goal)
V-JEPA 2.1 / 2-AC	Dense unlock; Franka CEM	CLEAR prior packs; not cited here; parallel
PLDM Sobal'25	E2E JEPA-WM multi-term VICReg	CLEAR closest foil
DINO-WM Zhou'25	Frozen DINOv2 WM + CEM	CLEAR foundation foil
LeJEPA / SIGReg (2511.08544)	Provable Gaussian anti-collapse	CLEAR inventum substrate
VICReg / Dreamer / TD-MPC / IRIS	Reg / generative foils	CLEAR
CEM / MPC / Push-T / OGBench	Planner + envs	CLEAR

This paper — CLEAR inventum

1. Claimed stable E2E JEPA from raw pixels with two-term objective (MSE + SIGReg).
2. Removes EMA, stop-grad, frozen encoder, PLDM's six+ weights → λ -only tuning.
3. Compact 15M AC WM; ~48× planning vs DINO-WM; Push-T **96.0±2.83**.
4. Physics suite: probing + VoE + emergent temporal straightening.
5. Open code `lucas-maes/le-wm`.

Subsequent — CLEAR / TENTATIVE

Work	Relation	Label
<code>lucas-maes/le-wm</code> + <code>le-wm.github.io</code>	Repro train/plan	CLEAR
Meta I-JEPA → ... → 2-AC chain	Remains CLEAR robotics path (do not demote)	CLEAR parallel
BEHAVIOR / real Franka with LeWM	Not shown	TENTATIVE
SIGReg E2E + V-JEPA 2.1 dense features	Open merge	TENTATIVE
π_0 / OpenVLA / GR00T	No LeWM cites; SigLIP/DINO towers	Architecture rhyme only
Hybrid LeWM+FM	Research proposal	TENTATIVE — not in-paper

Genealogy (CLEAR framing)

```
Predictive coding / LeCun AMI / JEPA abstract
→ I-JEPA (Assran 2023) – image multi-block latent prediction
  [pack: /workspace/ijepa-quiet/]
→ V-JEPA → V-JEPA 2 → 2.1 → 2-AC (Franka CEM)
  [pack: /workspace/vjepa21-quiet/] ← PARALLEL Meta line

Control-domain E2E JEPA-WM:
  PLDM (Sobal 2025) – multi-term VICReg E2E from pixels
  → **LeWM (this, 2026)** – MSE + SIGReg; compact latent CEM
    → optional 2026 BEHAVIOR local WM brick (Visionary TENTATIVE)
```

Archaeologist exam bite: "LeWM (Maes'26) = sibling control E2E JEPA-WM (PLDM → LeWM), not Meta V-JEPA child; inventum = SIGReg two-term stability + **0.98 vs 47 s** planning; Push-T **96±2.83**; **sim-only / no Franka**; keep Meta 2-AC CLEAR for real robot."

6.3 Stakeholder — produced despite empty seat

Buy / why schedule this for “Accompanying Models”:

- Completes JEPA arc: I-JEPA (idea) → V-JEPA 2.1/2-AC (scale + real robot) → **LeWM (tiny AC WM without brittle heuristics)**.
- Answers student pain: “JEPA collapses / needs EMA / six loss knobs.”
- Systems hook: **0.98 s vs 47 s**; **15M** on one GPU — syllabus-friendly vs Meta-scale.
- Clean three-way map (Fig. 2): E2E vs foundation-freeze vs reward/recon WMs.
- Open code + teachable Alg. 3.

Risks / costs / limits:

- **Sim-only** — no Franka/BEHAVIOR; do **not** replace V-JEPA 2-AC robotics lecture.
- Loses on Two-Room; slightly trails DINO on Cube — honesty required.
- Short horizon (H=5 / 25 steps); hierarchical WM admitted as future.
- Needs action-labeled offline coverage; low diversity hurts SIGReg.
- Empiricist seat weighted — expect deep number grilling; **Kanta not seated**.

Decision ask: Required companion after I-JEPA + V-JEPA 2-AC for “training stability & planning latency”; pair with PLDM/DINO-WM as foils. Optional if track only covers Meta SSL → Franka.

Stakeholder exam bite: "Buy LeWM as the **teachable stable small-WM recipe** + sub-second CEM story; don't buy it as a Franka/BEHAVIOR result or as a silent V-JEPA 2-AC replacement."

6.4 Scientific Reviewer — produced despite empty seat

Claims under review

1. Two-term MSE+SIGReg yields **stable E2E JEPA from pixels** without EMA/SG/freeze/multi-term VICReg.
2. Tunable loss HPs collapse to λ ; robust band enables bisection.
3. Compact latent enables **~48×** faster planning than foundation WMs at competitive / better control.
4. Latents encode physical structure (probes + VoE).

Strengths

1. Clean inventum: collapse problem → two-term objective → measurable stability + speed.
2. Fair foils: E2E (PLDM), foundation (DINO-WM), GC-RL/BC.
3. Ablations isolate thesis (recon hurts; λ band; M inert; dropout 0.1).
4. Physics suite beyond reward hacks; candid limitations (TwoRoom, short H, action labels).
5. Tab. 5 3-seed variance; Ablation Bot corpus rich.

Weaknesses / ask-for-revision

1. “First stable E2E JEPA” depends on definitional boundaries vs PLDM/LeJEPA.

- 2. Fig. 6 other-env numbers chart-heavy — want full numeric table.
- 3. CEM 300×30 dominates wall-clock; matched-iteration budgets beyond Fig. 3 needed.
- 4. Two-Room failure needs a fix (adaptive λ /dim), not only diagnosis.
- 5. **No real-robot transfer** — reviewers will ask for Franka/Bridge smoke.
- 6. Dropout is still a heuristic; BN projector / AdaLN zero-init unablated.
- 7. EMA/SG removal claimed but **not** factorial within LeWM.

Verdict: Strong methods + systems paper for the **small E2E control-WM** niche. Accept for Accompanying Models as **required companion** after I-JEPA/V-JEPA 2-AC; when citing robots, demand **V-JEPA 2-AC**, not LeWM; when citing stable E2E recipe, cite **LeWM / SIGReg**.

Reviewer exam bite: Demand Tab. 5 + Fig. 3 + Fig. 16 + Tab. 7 before slogan; cite Push-T **96.0±2.83** not bare “+18%”; punish Franka-as-result and BEHAVIOR-hybrid-as-result; fence TwoRoom caveat.

6.5 Visionary — produced despite empty seat (BEHAVIOR recipe; no Discord/Drive)

BEHAVIOR connection:
2025/'26 BEHAVIOR winners ride $\pi_{0.5}$ + **flow-matching** + **SigLIP/PaliGemma** — generative motor policies. LeWM shares the **planning interface** V-JEPA 2-AC popularized (optimize actions so imagined latents match a **goal image**) and adds **training stability** + **real-time latent MPC** on one GPU — attractive as a **local world-model module** beside a VLA, **not** the whole stack. **Do not invent** LeWM robot/BEHAVIOR scores — paper is Push-T / Cube / TwoRoom / Reacher only.

Proposed hybrid (TENTATIVE — do NOT invent as existing / in-paper result): $\pi_{0.5}$ /FM executes short-horizon motor chunks; **LeWM-style latent CEM** scores long-horizon feasibility / image-goal subgoals / VoE surprise as anomaly. Alt: V-JEPA 2.1 dense state → **LeWM-sized SIGReg** predictor for fast MPC.

BEHAVIOR need	LeWM-shaped response
Visual trunk	Default stays SigLIP-class ; LeWM = local WM / planner ablation
Real-time MPC Hz	Compact [CLS] CEM (cite 0.98 vs 47 s carefully — sim)
Training stability	E2E MSE+SIGReg; refuse EMA/SG stacks unless ablated necessary
Goal-image planning	Same CEM interface as 2-AC / DINO-WM
Language instructions	Keep SigLIP parallel — LeWM unlabeled
Low-diversity subtasks	Watch TwoRoom moral — adapt λ / prior
Eval honesty	Citing Push-T 96% ≠ BEHAVIOR q-score

CLEAR: Stable 2-term E2E JEPA + compact latent CEM is a viable small-WM path.
TENTATIVE: Official BEHAVIOR'26 baseline ships LeWM brick; hybrid LeWM-rollout + FM wins board.

2026 BEHAVIOR world-model / planning recipe (8 bets)

- 1. **Default WM recipe:** E2E JEPA + **single anti-collapse** (SIGReg or successor) — refuse EMA/SG unless RC1 says necessary.
- 2. **λ card:** publish λ (or bisection log) like latency cards — one-number honesty.
- 3. **Plan in compact latent:** prefer CLS/pooled for MPC frequency; keep dense features (V-JEPA 2.1) for perception probes.
- 4. **Solver card:** CEM vs first-order (Tab. 10) must be reported.
- 5. **Env complexity gate:** SIGReg Gaussian priors can hurt low-diversity household sub-tasks (TwoRoom).
- 6. **Physics unit tests:** VoE teleport/color + pose probes as CI before rollouts.
- 7. **Compose stacks:** LeWM imagination × $\pi_{0.5}$ /OpenPI/DP executors; tag `wm=lewm_sigreg`, `planner=cem_h5`, `pixels_only=true`.
- 8. **Honest non-goals:** course smoke ≠ 4-env paper SR; BEHAVIOR needs scene diversity beyond PushT — this pack = **WM stability + plan speed**, not demos.

Follow-up research

- 1. SIGReg-E2E on BEHAVIOR-1K / OXE play — planning Hz vs DINO-WM.
- 2. Image-goal CEM in BEHAVIOR scenes vs language-goal VLA.
- 3. VoE surprise as **safety monitor** for contact-rich skills.
- 4. Hierarchical LeWM for multi-room tasks (paper limitation).
- 5. Distill V-JEPA 2.1 → 15M LeWM encoder for edge robots.
- 6. Adaptive SIGReg for low-intrinsic-dim regimes (TwoRoom fix).

New applications (beyond Discord/Drive)

- Real-time latent MPC for tabletop rearrange with image goals.
- Offline “what-if” policy evaluation inside compact JEPA imagination.
- Physics-violation alarms for teleop assist.
- Teaching artifact: smallest complete JEPA-WM you can train overnight.

Visionary one-liner: LeWM is the **fast, stable, teachable JEPA world-model brick** — pair with V-JEPA 2-AC (real robot) and $\pi_{0.5}$ (motor skill); **hybrid is proposal, not result**.

Visionary exam bite: "For BEHAVIOR, default **SigLIP + FM**; ablate **LeWM-style** local SIGReg WM via Empiricist RC1–RC3; cite Push-T / **0.98 vs 47** as sim morals only; **do not invent** LeWM+FM hybrid as in-paper — **no Discord/Drive**."

(No Discord / Drive checklist content.)

7. Empiricist run pack (env / scripts / GPU — from DELIVERABLE)

Env setup

```
export LEWM_WORK_ROOT=$SCRATCH/lewm-smoke
cd /workspace/leworldmodel-quiet
bash env_outline/setup_mamba.sh
# mamba activate lewm-scout
mkdir -p logs runs checkpoints configs
# EDIT: partition/account when fleshing slurm_*.sh
# Pin lucas-maes/le-wm commit → logs/pins.txt
# Outlines only – do NOT sbatch from quiet box
```

File	Role
env_outline/conda_env.yaml	lewm-scout sketch
env_outline/setup_mamba.sh	Create env + dirs
env_outline/run_checklist.md	Ordered RC0→RC7 checklist
env_outline/README.md	Index + honesty; siblings
env_outline/slurm_*.sh	Named in DELIVERABLE §3–4 / README (RC1–RC7 cards) — flesh from run cards before cluster

GPU / RAM / TIME budget honesty

Assumptions: ViT-Tiny/S smoke, image ≤ 224 (prefer 128), batch 32–128, PushT-lite / synthetic — **not** full 4-env 10-epoch paper protocol.

Workload	GPU VRAM (peak)	Host RAM	Wall time (order)	Smoke
RC0 ingest	—	8–16 GB	<0.5 h CPU	Y
RC1 heuristics ×2–4	10–20 GB	32–64 GB	4–12 h	Y
RC2 λ ×4	10–20 GB	32 GB	3–10 h	Y
RC3 planning speed (infer)	ckpt	16–32 GB	≤1–4 h	Y
RC4 baselines ×3	12–24 GB	64 GB	6–16 h	Y / Partial
RC5 decoder ×2	10–20 GB	32 GB	2–6 h	Y
RC6 dropout/size	10–20 GB	32 GB	2–12 h	Partial
RC7 VoE/probe	4–12 GB	16–32 GB	1–4 h	Partial
Paper ~15M, 1 GPU, few hours, 10 ep ×4 envs	~12–24 GB	—	Paper claim — still larger than course matrix	N (cite)
Full DINO-WM foundation + Tab. 5 3-seed	multi-day	—	Out of course	N

Honesty rule: log

(arch, recipe, lambda, M, dropout, frames, res, steps, BS, #tokens, ms_encode, ms_cem, proxy_sr, wall_s, mem) . Cut resolution and batch before lying about budgets. Never amortize paper 96% / 0.98 s / 48× / 90 onto a Partial stub.

No Franka.

Course envelope: target $\leq$ ~40–60 GPU-h for P0 matrix (RC0–RC4 + RC5). Full App. G × 3 seeds × 4 envs = well above course.

Success criteria vs paper claims

Paper claim	Minimal course criterion	Not required
Stable E2E JEPA; no SG/EMA/pretrained	RC1: sigreg trains; noreg collapses	Fig. 18 exact curves
Tunable HPs 6 → 1 (λ)	RC2: mid- λ best	Exact bisection demo
Plans ~48× faster than foundation WM	RC3: compact $\ll$ DINO-stub wall-time	Exact 0.98 / 48×
Push-T +18% vs PLDM; competitive vs DINO	RC4: lewm $\geq$ pldm_lite; ~dino	96 / 78 / 92 absolutes
No decoder better (Tab. 7)	RC5 directional	Exact 96 vs 86
Dropout 0.1 sweet spot	RC6a	Full {0, 0.1, 0.2, 0.5}
Physical probes + VoE	RC7 stubs	Tab. 1 / Fig. 8 significance
TwoRoom / Cube / Reacher full	Cite + optional 1 lite env	Full Fig. 6 4-env
Franka / BEHAVIOR	Cite absence	Inventing numbers

Pass: Empiricist executes RC1 + RC2 + RC3 (+ RC4 if GPU) with paper-aligned directional rankings + GPU honesty logged.

Fail / overclaim: "reproduced Maes'26 / matched 96% / verified 48× / Franka / BEHAVIOR win" from smoke alone.

8. Slide Maker brief

Role	Weight / timing guide	Focus
Empiricist	3 · LONGEST · ~16–18	H-stable/ λ /speed/rank; RC1–RC4; GPU honesty; must-cite tattoo; exam bite
Archaeologist	empty seat · ~12–14	SIBLING PLDM → LeWM vs Meta I-JEPA → 2-AC; sim-only fence
Stakeholder	empty seat · ~12–14	Buy stability+latency module; risks (sim-only, TwoRoom); decision ask
Reviewer	empty seat · ~12–14	Tab.5+Fig.3+Fig.16+Tab.7; verdict Accept companion; no Franka-as-LeWM
Visionary	empty seat · ~12–14	BEHAVIOR SigLIP+FM default + LeWM local WM ablation; hybrid=TENTATIVE; no Discord/Drive

Spine (paste into Slide Maker) — 5 bullets:

- 1. Stable ~15M E2E JEPA WM from pixels — MSE + SIGReg only; drops EMA / stop-grad / frozen / multi-VICReg.
- 2. Must-cite: Push-T 96.0±2.83; latency 0.98 vs 47 (~48×); fixed-FLOP 90 vs 13; recon 96 → 86; $\lambda \in [0.01, 0.2]$ / cliff 0.5; TwoRoom caveat; no Franka.
- 3. Taxonomy: DINO-WM (freeze) | Dreamer/IRIS (recon/reward) | Meta I-JEPA → ... → 2-AC (parallel) | PLDM → LeWM (this).
- 4. Lineage (Empiricist=3 Nykolas LONGEST): sibling control recipe PLDM → LeWM, not Meta V-JEPA descendant; inventum = heuristics-off stability + compact latent CEM speed.
- 5. BEHAVIOR'26 + honesty: default SigLIP+FM; LeWM = local WM ablation; Empiricist RC1–RC4 directional only; do not invent LeWM+FM hybrid as in-paper — no Discord/Drive.

GitHub title sketch: [ECE 605] - LeWorldModel Empiricist Making Three

Full Slide Maker brief: /workspace/leworldmodel-quiet/SLIDE_BRIEF.md

Paths

Artifact	Path
This pack	/workspace/leworldmodel-quiet/KANTA_PACK.md
PDF	/workspace/leworldmodel-quiet/KANTA_PACK.pdf
Slide brief	/workspace/leworldmodel-quiet/SLIDE_BRIEF.md
Senior Research	/workspace/leworldmodel-quiet/DELIVERABLE.md
Summarizer	/workspace/papers/leworldmodel-2603.19312-summary.md
Ablations	/workspace/papers/leworldmodel-2603.19312-ablations.md (AUTHORITATIVE)
Paper notes	/workspace/leworldmodel-quiet/paper_notes.md
Env outlines	/workspace/leworldmodel-quiet/env_outline/
Paper PDF	/workspace/papers/leworldmodel-2603.19312.pdf
Page figs	/workspace/papers/leworldmodel-figs/page-*.png
Lineage siblings	/workspace/ijepa-quiet/ · /workspace/vjepa21-quiet/
Code / site	https://github.com/lucas-maes/le-wm · https://le-wm.github.io

Pack synthesized 2026-09-29 PT (HST) for KanBot / ECE 605 — Dec 2 Accompanying Models · Empiricist=3 (Nykolas) LONGEST · Kanta NOT seated · all five produced · Ablation Bot AUTHORITATIVE · sim-only / no Franka · SIBLING PLDM → LeWM ≠ Meta 2-AC · no Discord/Drive · no invented BEHAVIOR hybrid as in-paper.